

Course No.: Math-1203
 Full Marks: 72

Course Title: Vector, Matrix & Ordinary Differential equations
 Time: Three Hours

- N.B.: i) Answers Six question taking Three from each section.
 ii) Figures in the margin indicate full marks.
 iii) Use separate answer script for each section.

SECTION-A

- Q.1(a) Determine whether the following vectors are linearly dependent or independent? 04
 $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} - 4\hat{k}$, $\vec{c} = 4\hat{i} + 3\hat{j} - \hat{k}$
 (b) Find the angles which the vector $\vec{A} = \hat{i} - 6\hat{j} + 2\hat{k}$ makes with coordinate axes. 04
 (c) If $F = (xz^2y - z)\hat{i} + (xz^3 + y^2)\hat{j} + 2xz^2z^2\hat{k}$, then find $\nabla(\nabla \cdot F)$ at the point (2, -1, 0) 04
 Q.2(a) What is the physical significance of $\nabla \cdot \vec{v}$? 04
 (b) If $\vec{r} = 3xy\hat{i} - y^2\hat{j}$, then evaluate $\int_C \vec{r} \cdot d\vec{r}$ where C is the curve in the xy plane, $y = 2x^2$ from (0, 0) to (1, 2) 04
 (c) Evaluate $\iint_S \vec{A} \cdot \hat{n} dV$ for $\vec{A} = y\hat{i} + 2z\hat{j} - x\hat{k}$ and S is the surface of the plane $2x + y = 6$ in the first octant cut off the plane $z=4$. 04

Q.3 State and prove the divergence theorem. Also verify this theorem considering a suitable example. 12

Q.4(a) Find the value of a, if 04

$$[a \ 4 \ 1] \cdot \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} a \\ 4 \\ 1 \end{bmatrix} = 0 \quad \text{Where 0 is } |0| \text{ null matrix.}$$

(b) Find the rank of the matrix $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ 2 & 2 & 5 & 0 \end{bmatrix}$ by reducing to normal form. 04

(c) Define identity matrix and symmetric matrix. 04

SECTION-B

Q.5(a) What is the inverse of a matrix? 06

(b) Find the inverse of matrix 06

$$\begin{bmatrix} 1 & 2 \\ 5 & 11 \end{bmatrix} \text{ by elementary transformation.}$$

Q.6(a) Define exact differential equation. 02

(b) Show that the necessary condition for a first order differential equation $Mdx + Ndy = 0$ to be an exact differential equation is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. 05

(c) Solve: $(2x - y + 1)dx + (2y - x - 1)dy = 0$. 05

Q.7(a) Solve the differential equation $\frac{dy}{dx} + y = e^{-x}$. 06

(b) Solve $\frac{d^2y}{dx^2} + y = \sin x$. 06

Q.8(a) Derive the pendulum equation. Solve it for the case of small oscillation. 06

(b) Solve $\frac{dy}{dx} + y = e^x$, $y(0) = 1$. 06

- N.B.: i) Answer any Six questions, taking Three from each section.
ii) Figures in the margin indicate full marks.
iii) Use separate answer script for each section.

SECTION-A

- Q.1(a) Define the following terms:
Square matrix, Diagonal matrix, Upper triangular matrix, Symmetric matrix, Hermitian matrix. 03
- (b) Prove that multiplication of matrices is distributive with respect to matrix addition. 04
- (c) What is transpose of a matrix? For any two matrix A and B, Prove that $(AB)^T = B^T A^T$. 05
- Q.2(a) Find the value of x, y, z and a which satisfy the matrix equation 03
- $$\begin{pmatrix} x+3 & 2y+x \\ z-1 & 4a-6 \end{pmatrix} = \begin{pmatrix} 0 & -7 \\ 3 & 2a \end{pmatrix}$$
- (b) Prove that a square matrix A to possess an inverse is that matrix A is non-singular. 04
- (c) Find the rank of the matrix 05

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{pmatrix}$$

- Q.3(a) Find the inverse of $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$ 04
- (b) Determine the values of α, β, γ when $\begin{pmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{pmatrix}$ is orthogonal. 04
- (c) Show that the diagonal elements of a Hermitian matrix are real and of a skew Hermitian matrix are either zeros or pure imaginary. 04
- Q.4(a) Show that the three vectors $\vec{A} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{B} = \hat{i} - 4\hat{k}$ and $\vec{C} = 4\hat{i} + 3\hat{j} - \hat{k}$ are linearly dependent. 05
- Determine a relation between them and hence show that the terminal points are collinear.
- (b) Find the angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point (1, -2, 1). 04
- (c) Find the constants a, b, c so that the vector $\vec{V} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational. 03

SECTION-B

- Q.5(a) Solve: $(xy + y^2)dx + (x^2y - x^2)dy = 0$ 03
- (b) Define integrating factor. Find the integrating factor of the differential equation 05
- $$\frac{dy}{dx} + y = 3x^2 - 3x$$
- Hence solve it. 04
- (c) Find the general solution of 12
- $$\frac{dy}{dx} + \frac{y}{2x} = \frac{1}{y^3}$$
- when $y(1) = 2$.
- Q.6 Solve the following differential equations:
- (i) $(D^2 + 9)y = x^2 + x + 1$
- (ii) $(D^2 - 4D + 4)y = e^x + \cos 2x + x^2$
- (iii) $y'' - 4y' + 3y = 3e^x \cos x + 2xe^{3x}$
- Q.7(a) Solve the differential equation $\frac{d^2y}{dx^2} + y = \tan x$ 06
- (b) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$ 06
- Q.8(a) A parachutist is falling with speed 55ms^{-1} when his parachute opens. If the air resistance is $WV^2/25$ N, where W is the total weight of the man and parachute, find his speed as a function of the time t after the parachute opened. 06
- (b) One of the basic equations in electric circuits is 06
- $$L \frac{di}{dt} + Ri = E(t)$$
- Where L (henries) is called the inductance, R (Ohms) the resistance, i (amperes) the current and E (volts) the electromotive force or emf. Find the current when $E(t) = E_0$, and the initial current is i_0 , R and L are constants.

- N.B. (i) Answer any Six questions, taking Three from each section
(ii) Figures in the margin indicate full marks
(iii) Use separate answer script for each section

SECTION-A

- Q.1(a) Write short note on the followings: Skimming, Scanning, SQ3R technique. 06
(b) Convert the form of the words as directed and make sentences with new formation: Friend(adjective), Threat(verb), Danger(verb), Know(noun), Wise(Noun), Form(adjective) 06
- Q.2(a) Write RP consonants with examples. 06
(b) Transcribe the following sentence using IPA: Mechatronics is the newly established department of RUET. 04
(c) Define: Vowels, Diphthongs 02
- Q.3(a) Write the meaning of the following idioms: Devil's advocate, In a fix, Last straw, Piece of cake. 04
(b) Write the usage of the following punctuation: Comma, Colon, Hyphen, Apostrophe 08
- Q.4(a) Write an argumentative paragraph on the following: "In democracies, servants are, in some sort, equal to their masters." 08
(b) Write short note on the followings: Topic sentence, Thesis statement 04

SECTION-B

- Q.5(a) Mention different parts of a business letter. 06
(b) Write a complain letter to an electric company for supplying goods in delay and its poor quality. 06
- Q.6(a) How to look up a dictionary Entry? 04
(b) How to write in-text citation of APA style? 04
(c) Write the reference list using APA style: i. One book with one author 04
ii. One book with three authors
iii. An Edited Book
iv. Dictionary entry
- Q.7(a) Discuss the parts of a Report. 06
(b) Write a job application along with CV. 06
- Q.8(a) Explain EDNA with Examples. 06
(b) Write a short Note on Prepositional phrase. 07
(c) Fill the gap and make sentence. 07
i) In association _____
ii) With a view _____
iii) _____ terms of _____
iv) Subject _____

Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

B.Sc. Engineering 1st Year Even Semester Examination 2016

Course No: Math 1203

Full Marks: 72

Course Title: Vector, Matrix and Ordinary Differential Equations

Time: Three Hours

N.B.: i) Answer any Six questions, taking Three from each section.

ii) Figures in the margin indicate full marks.

iii) Use separate answer script for each section.

SECTION-A

Q.1(a) What is matrix and orthogonal matrix? Prove that the matrix $\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ is orthogonal. 04

(b) If $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, then find A^2 and show that $A^2 = A^{-1}$. 04

(c) What is the Jacobian matrix? If $T(u, v) = (u^2 - v^2, 2uv)$, then 04

(i) Find the velocity of $u(t) = (t, t^2)$ when $t = 1$.

(ii) Find the Jacobian of $T(u, v)$.

Q.2(a) Reduce the matrix C to echelon form and hence find its rank. 04

$$C = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{pmatrix}$$

(b) Test the consistency of the following equations and then solve it. 04

$$x + y - z = 1$$

$$2x + 3y - 3z = 3$$

$$x - 3y + 3z = 2.$$

(c) Find the eigen values and corresponding eigen vectors of the matrix 04

$$D = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}.$$

Q.3(a) Find a unit vector perpendicular to the plane of $\vec{A} = 2\vec{i} - 6\vec{j} - 3\vec{k}$ and $\vec{B} = 4\vec{i} + 3\vec{j} - \vec{k}$. 04

(b) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 04

(c) If $\vec{V} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + az)\vec{k}$, determine the constant 'a' so that the divergence of $\vec{V}$ is zero. 04

Q.4(a) Verify Green's theorem for $\oint_c \{ (xy - x^2)dx + x^2ydy \}$ where c is the perimeter of the triangular area bounded by the lines $x=1$, $y=0$ and $y=x$. 06

(b) If $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$, evaluate $\iint_S \vec{F} \cdot \vec{n} ds$ where S is surface of the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$, $z=1$. 06

SECTION-B

- Q.5(a) Solve $(x+y)^2 \frac{dy}{dx} = a^2$. 04
- (b) Define exact differential equation. Solve $y(1+xy)dx = x(1-xy)dy = 0$. 04
- (c) Find the singular solution and complete solution of the differential equation $y = px + p - p^2$ where $p = \frac{dy}{dx}$. 04
- Q.6(a) What is differential equation? Mention three applications of differential equation. 04
- (b) Obtain the differential equation associated with the primitive $x^2 y^3 + x^3 y^5 = C$. 04
- (c) Solve $\frac{dy}{dx} + \frac{1}{3}y = \frac{1}{3}(1-2x)y^4$, $y(0) = 1$. 04
- Q.7 Find the general solution of the following differential equations: 03
- (a) $(D^2 - 9D + 18)y = e^{-3x}$ 04
- (b) $(D^2 - 2D + 3)y = x^3 + \sin x$ 05
- (c) $(D^2 + 4)y = x^2 \sin 2x$
- Q.8(a) If the population of a country doubles in 50 years, in how many years will it treble under the assumption that the rate of increase is proportional to the number of inhabitants? 04
- (b) A body of mass m kg falls from rest in a medium for which the resistance (N) proportional to the square of the velocity (ms^{-1}). If the terminal velocity is 50 ms^{-1} , then find the velocity at the end of 2 seconds. 04

Course No: Math 1227

Course Title: Vector, Matrix and Ordinary Differential Equation

Full Marks: 72

Time: Three Hours

N.B. i) Answer any Six questions, taking Three from each section.

ii) Figures in the margin indicate full marks.

iii) Use separate answer script for each section.

$$A' = P$$

SECTION-A

$$AA' = I$$

$$A^{-1} = I$$

Q.1(a) Define Symmetric, Transpose, Orthogonal and Idempotent matrices. 04

(b) Prove that $(AB)^{-1} = B^{-1}A^{-1}$ where A and B are two non-singular matrix. 04

(c) Find the inverse of the matrix $A = \begin{bmatrix} 3 & -2 & 2 \\ 1 & 2 & -3 \\ 4 & 1 & 2 \end{bmatrix}$ 04

Q.2(a) Find the rank of the matrix $A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$ 06

(b) Solve the following linear system by a matrix formula. 06

$$x + 2y + z = 0$$

$$2x - y - z = 2$$

$$5x + 7y - z = -3$$

Q.3(a) Show that $\vec{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is Irrotational. Find ϕ such that $\vec{A} = \vec{\nabla} \phi$. 04

(b) Find the directional derivatives of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$ 04

(c) If $\vec{F} = (5xy - 6x^2)\hat{i} + (2y - 4x)\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the curve C in the xy plane $y = x^3$ from the point $(1, 1)$ to $(2, 8)$. 04

Q.4(a) Apply the Divergence theorem to compute $\int \vec{u} \cdot \hat{n} ds$ where s is the surface of the cylinder $x^2 + y^2 = a^2$ bounded by the planes $z = 0, z = h$ and where $\vec{u} = \hat{i}x - \hat{j}y + \hat{k}z$. 06

(b) If $\vec{F} = [2xz, -x, y^2]$ and V is the region bounded by the surfaces $x = 0, y = 0, y = 6, z = x^2, z = 4$, then show that $\iiint_V \vec{F} \cdot d\vec{v} = [128, -24, 384]$. 06

SECTION-B

Q.5(a) Form a differential equation from the relation $Ax^2 + By^2 = 1$. 04

(b) Solve: $\frac{dy}{dx} = \frac{2x + y - 2}{3x + y - 3}$. 05

(c) Find the general solution of $xy = (y^2 - 1)dx$. 03

Q.6(a) Solve: $\frac{dy}{dx} + 2y = x$, $y(0) = 0$

06

- (b) What is an integrating factor? Find an integrating factor of $ydx + (3 + 3x - y)dy = 0$. Hence solve it.

06

Q.7 Solve the following differential equations

12

(a) $(D^2 + 1)y = x$

(b) $(D^3 - 2D^2 + 4D)y = e^x \cos x$

(c) $(D^2 - D + 1)y = \sin 2x$ where $D = \frac{d}{dx}$

B.D.J. 69

Q.8(a) Using the method of variation of parameter solve the differential equation

06

$$\frac{d^3 y}{dx^3} + 2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} - 2y = e^x + x^2$$

- (b) In an electric circuit contains a resistance R ohms and a condenser of capacitance C (farads) in series and an emf E (volts). The charge q (coulombs) on the condenser is

06

given by: $R \frac{dq}{dt} + \frac{q}{C} = E$

If $R = 10$ ohms, $C = 10^{-2}$ farad and $E(t) = e^{i(3\pi t - 120\pi t)}$ volts, then find q assuming that $q = 0$ when $t = 0$.

N.B. i) Answer any **Six** questions, taking **Three** from each section.

ii) Figures in the margin indicate full marks.

iii) Use separate answer script for each section.

SECTION-A

Q.1(a) What is an orthogonal matrix? Show that the matrix $\begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$ is orthogonal. 04

(b) Determine by reducing to normal form the rank of matrix $A = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ 2 & 2 & 8 & 0 \end{pmatrix}$. 05

(c) If a matrix has an inverse, then show that the inverse is unique. 03

Q.2(a) Apply the test of rank to examine if the following equations 06

$$2x - y + 3z = 8$$

$$-x + 2y + z = 4$$

$$3x + y - 4z = 0$$

are consistent and if consistent, solve them.

(b) Find the characteristic roots and characteristic vectors for the matrix $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. 06

Q.3(a) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\underline{i} - 3\underline{j} + 2\underline{k}$. 04

(b) Prove that 04

i. $\vec{r} \times (\vec{r} \cdot \nabla \phi) = 0$

ii. $\vec{r} \cdot (\vec{r} \times \vec{A}) = 0$

(c) Find the equations of the tangent plane and the normal line to the surface $xyz^2 = 4$ at $(2, 2, 1)$. 04

Q.4(a) If $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, evaluate $\iint_S \vec{F} \cdot \hat{n} dS$, where S is the surface of the cube bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$. 06

(b) State and prove Stoke's theorem. 06

SECTION-B

Q.5(a) Obtain the differential equation corresponding to $y^2 = 4a(x + a)$. 04

(b) If $M(x, y)dx + N(x, y)dy = 0$ is a homogeneous equation, then show that the change of variables $y = vx$ transform the homogeneous equation into a separable equation in the variables v and x . 04

(c) Solve: $x^3 dx + (y + 1)^2 dy = 0$, satisfying the initial conditions $x = 1$, $y = 2$. 04

(a) Solve: $L \frac{di}{dt} + Ri = E \sin 2t$, where L, R, E are constants, subjected to the condition $i = 0$ when $t = 0$. 06

(b) Find the general solution of $\frac{dy}{dx} + \frac{1}{3}y = \frac{1}{3}(1 - 2x)y^4$, with $y(0) = 1$. 06

Q.7 Solve the following differential equations:

12

i. $(D^2 - 5D + 6)y = e^{3x} + xe^{2x}$

ii. $(D^3 - 2D + 4)y = e^x \cos x$

iii. $(D^2 - D + 1)y = \sin 2x$

where $D = \frac{d}{dx}$.

8(a) A body executes damped forced vibrations that are governed by the differential equation $\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + b^2x = e^{-kt} \sin \omega t$ where all the symbols have their usual meanings. 06

Solve the above differential equation for both the cases where $\omega^2 \neq (b^2 - k^2)$ and $\omega^2 = (b^2 - k^2)$.

~~(b)~~ Apply the method of variation of parameters to solve

06

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$$

- N.B. i) Answer any Six questions, taking Three from each section.
ii) Figures in the margin indicate full marks.
iii) Use separate answer script for each section.

SECTION-A

- Q.1(a) Show that the three vectors $\vec{A} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{B} = \hat{i} - 4\hat{k}$ and $\vec{C} = 4\hat{i} + 3\hat{j} - \hat{k}$ are linearly dependent. Determine a relationship between them and hence show that the terminal points are collinear. 04
- (b) Find a unit tangent vector at the point where $t = 2$ on the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - 6t$. 04
- (c) Show that $\vec{E} = \frac{\vec{r}}{r^2}$ is irrotational. Find ϕ such that $\vec{E} = -\nabla\phi$ and such that $\phi(a) = 0$ where $a > 0$. 04
- Q.2(a) If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4xz\hat{k}$, evaluate $\iiint_V (\nabla \times \vec{F}) dV$, where V is the closed region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$. 06
- (b) Evaluate $\iint_S \vec{A} \cdot \hat{n} dS$ where $\vec{A} = y\hat{i} + 2xz\hat{j} - x\hat{k}$ and S is the surface of the plane $2x + y = 6$ in the first octant cut off by the plane $x = 4$. 06
- Q.3(a) What is an inverse of a matrix? Find the inverse of the matrix 06
- $$\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$
- (b) Examine if the system of equations 06
- $$\begin{aligned} x + y + 4z &= 6 \\ 3x + 2y - 2z &= 9 \\ 5x + y + 2z &= 13 \end{aligned}$$
- is consistent. Find also the solution if the system is consistent.
- Q.4(a) Transform the matrix 06
- $$A = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{pmatrix}$$
- into normal form and hence find its rank.
- (b) Verify Cayley-Hamilton's theorem for the matrix 06
- $$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

SECTION-B

- Q.5(a) Define differential equation with example. From the differential equation corresponding to 04
- $$y = ae^{2x} + be^{-2x} + ce^x \text{ where } a, b, c \text{ are parameters.}$$
- (b) Solve: $(2x - y + 1)dx + (2y - x - 1)dy = 0$ 04
- (c) Find the general solution of $\frac{dy}{dx} + 2xy = 2e^{-x^2}$ 04
- Q.6(a) What is integrating factor? Find the integrating factor of $ydx + (3 + 3x - y)dy = 0$, and hence solve it. 04
- (b) Find the current from $L \frac{di}{dt} + Ri = E_0 \sin \omega t$ where L, R, E_0 are constants. 04
- (c) Find the general solution of $\frac{dy}{dx} = x^3 y^3 - xy$. 04
- Q.7(a) Solve the following differential equations: 12
- i) $(D^2 + 4D + 3)y = e^{-2x}$
- ii) $(D^2 - 9)y = 6e^{2x} + xe^{3x}$
- iii) $(D^2 + 2D + D)y = e^{2x} + x^3 + x$
- Q.8(a) Solve the following differential equation: 06
- $$y + 5y' + 4y'' = 16x + 20e^x; y(0) = 0, y'(0) = 3$$
- (b) By the method of variation of parameters, solve the differential equation 06
- $$\frac{d^2y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = \sin^2 x$$

- N.B. i) Answer any Six questions, taking Three from each section.
ii) Figures in the margin indicate full marks.
iii) Use separate answer script for each section.

SECTION-A

- Q.1(a) Determine whether the vectors $\vec{r}_1 = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{r}_3 = -2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{r}_4 = 3\hat{i} + 2\hat{j} + 5\hat{k}$ are linearly independent or dependent. 04
- (b) Determine the volume of a tetrahedron whose adjacent edges are $\vec{A} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{B} = \hat{i} - 4\hat{k}$, $\vec{C} = 4\hat{i} + 3\hat{j} - \hat{k}$. 04
- (c) Show that $\vec{F} = e^x \cos z \hat{i} + 2y\hat{j} - e^x \sin z \hat{k}$ is irrotational. Find a scalar function ϕ such that $\vec{F} = \nabla\phi$. 04

- Q.2(a) Show that $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is a conservative force field. Find the scalar potential for $\vec{F}$. 06
- (b) If $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. 06

- Q.3(a) State and prove divergence theorem. 06
- (b) Evaluate $\iiint_V (2x + y) dV$, where V is the closed region bounded by the cylinder $z = 4 - x^2$ and the planes $x = 0, y = 0, y = 2$ and $z = 0$. 06

- Q.4(a) What is orthogonal matrix? If A and B are two orthogonal matrices, then show that AB is also orthogonal matrix. 04
- (b) Express the matrix $\begin{pmatrix} -1 & 7 & 1 \\ 2 & 3 & 4 \\ 5 & 0 & 5 \end{pmatrix}$ as the sum of a symmetric and a skew-symmetric matrix. 04
- (c) Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. 04

SECTION-B

- Q.5(a) Define rank of a matrix. Reduce the matrix $A = \begin{pmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{pmatrix}$ to the normal form and hence find its rank. 06

- (b) Find the characteristics roots and characteristics vectors for the matrix $A = \begin{bmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix}$. 06

- Q.6(a) What is differential equation? Obtain the differential equation of the family of curves $xy = ae^x + be^{-x}$ where a and b are parameters. 04

- (b) Solve $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$. 04
- (c) Find the integrating factor of $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$. Hence solve it. 04

- Q.7 Solve the following differential equations: (i) $(D^2 + 2D + 1)y = e^x + e^{-x}$; (ii) $(D^2 + 1)y = xsinx$; (iii) $(D^2 - 4D + 4)y = x^2e^{2x}$; where $D = \frac{d}{dx}$. 12

- Q.8(a) Using the method variation of parameters, solve the differential equation $(x-1)D^2y - xDy + y = (x-1)^2$, where $D = \frac{d}{dx}$. 06

- (b) Solve $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = x$ subject to the condition that when $x = 1, y = 0$ and $\frac{dy}{dx} = 0$. 06

N.B. i) Answer any Six questions, taking Three from each section.

ii) Figures in the margin indicate full marks.

iii) Use separate answer script for each section.

iv) CO-Course Outcome.

SECTION-A

	Marks	CO
Q.1(a) Determine whether the vectors $\vec{r}_1 = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{r}_3 = -2\hat{i} + \hat{j} - 3\hat{k}$, and $\vec{r}_4 = 3\hat{i} + 2\hat{j} + 5\hat{k}$ are linearly dependent or independent.	03	CO1
(b) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction $2\hat{i} - 3\hat{j} + 6\hat{k}$.	03	CO1
(c) Show that $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ is conservative force field. Find a scalar potential ϕ such that $\vec{F} = \nabla\phi$.	04	CO1
Q.2(a) Find the work done in moving a particle once around a circle C in the xy plane if the circle has centre at the origin and radius z and the force field is given by $\vec{F} = y^3\hat{i} - x^3\hat{j}$.	05	CO1
(b) If $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$; and s is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, evaluate $\iint_S \vec{F} \cdot \hat{n} ds$.	05	CO1
Q.3(a) State and prove stoke's theorem.	05	CO1
(b) Verify the divergence theorem for $\vec{A} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by $x^2 + y^2 = 4$, $Z = 0$, $Z = 3$.	05	CO1
Q.4(a) What is a matrix? Show that every real symmetric matrix is Hermitian.	03	CO2
(b) What is an inverse of a matrix? Show that the inverse of a matrix is unique.	03	CO2
(c) Find adjoint and hence inverse of the matrix $\begin{bmatrix} 1 & 2 & 1 \\ 5 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$.	04	CO2

SECTION-B

Q.5(a) Define rank of a matrix. Reduce the matrix $A = \begin{pmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{pmatrix}$ into normal form and hence find the rank.	05	CO2
(b) Find the eigen values and the corresponding eigen vectors of the matrix $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.	05	CO2
Q.6(a) What is exact differential equation? Find the condition that a differential equation to be exact.	04	CO3
(b) Obtain the differential equation associated with the primitive $y = Cx^2 + C^2$.	03	CO3
(c) Solve $(x^2 + 1)\sin y dx + x \cos y dy = 0$	03	CO3
Q.7 Solve the following differential equations:		
(a) $(D^2 - 4D + 3)y = 3e^x \cos x + 2xe^{3x}$	04	CO3
(b) $(D^2 + 5D + 4)y = 3 - 2x$	03	CO3
(c) $(D^2 - 3D + 2)y = \sin 3x$	03	CO3
Q.8(a) Solve $(x^2 D^2 - 3xD + 4)y = x^2$; $y(1) = 0$, $y'(e) = e^2$ where $D = \frac{d}{dx}$.	05	CO3
(b) Use the method of variation of parameters solve the differential equation $\frac{d^3 y}{dx^3} + 2\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 2y = e^x + x^2$	05	CO3

N.B. i) Answer any Six questions, taking Three from each section.

ii) Figures in the margin indicate full marks.

iii) Use separate answer script for each section.

iv) CO- Course Outcome

SECTION-A		Marks	CO
Q.1(a)	Forces of magnitudes 5 and 3 units acting in the direction $6\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + 6\hat{k}$ respectively act on a particle which is displaced from the point (2, 2, -1) to (4, 3, 1). Find the work done by the forces.	03	CO1
(b)	In what direction from the point (1, 3, 2) is the directional derivative of $\phi = 2xz - y^2$ a maximum? What is the magnitude of this maximum?	03	CO1
(c)	Show that $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is a conservative force field. Also, find the scalar potential.	04	CO1
Q.2(a)	If $\vec{F} = (2x + y)\hat{i} + (3y - x)\hat{j}$, then evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve in the xy plane consisting of the straight lines from (0, 0) to (2, 0) and then to (3, 2).	05	CO1
(b)	Evaluate: $\iiint_S \phi \hat{n} dS$ where $\phi = \frac{3}{8}xyz$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.	05	CO1
Q.3(a)	State and prove divergence theorem.	05	CO1
(b)	Verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the boundary of the region defined by $y = \sqrt{x}$ and $y = x^2$.	05	CO1
Q.4(a)	Define symmetric and skew- Hermitian matrix with example.	02	CO2
(b)	Show that every square matrix can be uniquely expressed as the sum of a symmetric and skew- symmetric matrices.	03	CO2
(c)	Define rank of a matrix. Reduce the matrix	05	CO2
$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$		$\dots \frac{k}{2} + k \quad \frac{k}{2} + k \quad \frac{k}{2} + k \quad \frac{k}{2}$	
into normal form and hence find its rank.			

SECTION-B		Marks	CO
Q.5(a)	Determine for what values of λ and μ the system	05	CO2
$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ x + 2y + \lambda z &= \mu \end{aligned}$			
has (i) No solution,			
(ii) Unique solution			
(iii) Infinite number of solutions			
(b)	Find the Eigen values and the corresponding Eigen vectors of the following	05	CO2
matrix: $A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$		$6 - \lambda^2 \quad 3x^2 + 8x^2 - 3x - 8\sqrt{x}$	

Q.6(a) Define a differential equation. Give examples. Mention some applications of differential equations. 03 CO3

(b) Solve the initial value problem: $xsinydx + (x^2 + 1)cosydy = 0$, $y(0) = \frac{\pi}{2}$. 04 CO3

(c) Find the constant A such that the equation $\left(\frac{Ay}{x^3} + \frac{y}{x^2}\right)dx + \left(\frac{1}{x^2} - \frac{1}{x}\right)dy = 0$ is exact and solve the resulting equation. 03 CO3

Q.7(a) Identify and solve the first order differential equation: 05 CO3

$$xy - \frac{dy}{dx} = y^3 e^{-x^2}.$$

(b) Find the complete solution of $y'' - 6y' + 25y = 0$; $y(0) = -3$, $y'(0) = -1$. 05 CO3

Q.8 Solve completely the following differential equations:

(a) $(D^2 - 4)y = e^x - \sin x$ 03 CO3

(b) $(D^2 - 4D + 4)y = x^2 + e^x$ 03 CO3

(c) $(D^2 + 1)y = x \sin x$ 04 CO3

- N.B. i) Answer any Six questions, taking Three from each section.
ii) Figures in the margin indicate full marks.
iii) Use separate answer script for each section.
iv) CO-Course Outcome.

SECTION-A

- | | Marks | CO |
|--|-------|-----|
| Q.1(a) Are the vectors $\vec{A} = 2\vec{i} - \vec{k}$, $\vec{B} = 3\vec{i} - 2\vec{j} + \vec{k}$, and $\vec{C} = 3\vec{i} + 2\vec{j} - \vec{k}$ linearly independent? if not, find a dependent relation between them. | 04 | CO1 |
| (b) Find the unit tangent vector at the point where $t = 2$ on the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - 6t$. | 03 | CO1 |
| (c) Find the curvature k for the curve $x = t$, $y = t^2$, $z = \frac{2t^3}{3}$. | 03 | CO1 |
| Q.2(a) Write down the geometrical interpretation of $\vec{\nabla} \Phi$. | 05 | CO1 |
| (b) Evaluate $\int_C \vec{A} \cdot d\vec{r}$ along the curve $x^2 + y^2 = 1$, $z = 1$, in the positive direction from $(0, 1, 1)$ to $(1, 0, 1)$ if $\vec{A} = (yz + zx)\vec{i} + xz\vec{j} + (xy + 2z)\vec{k}$ | 05 | CO1 |
| Q.3(a) Evaluate $\oint_S \Phi \hat{n} ds$ where $\Phi = 4x + 3y - 2z$ and S is the surface of $2x + y + 2z = 6$ bounded by $x = 0$, $x = 1$, $y = 0$ and $y = 2$ | 05 | CO1 |
| (b) Verify Stokes theorem for $\vec{A} = 3y\vec{i} - xz\vec{j} + yz^2\vec{k}$ where s is the surface of the paraboloid $2z = x^2 + y^2$ bounded by $z = 2$ and c is its boundary traversed in the clockwise direction. | 05 | CO1 |
| Q.4(a) Define matrix and symmetric matrix. If A and B are n -squared symmetric matrices then prove that AB is symmetric if and only if A and B are commute matrices. | 04 | CO2 |
| (b) Given that $A = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix}$ then prove that $A^n = \begin{bmatrix} 1 + 10n & -25n \\ 4n & 1 - 10n \end{bmatrix}$ | 03 | CO2 |
| (c) Show that the matrix $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ is orthogonal. | 03 | CO2 |

SECTION-B

- | | | |
|---|----|-----|
| Q.5(a) For what values of λ , the system: $\begin{aligned} x + y + z &= 1 \\ x + 2y + 4z &= \lambda \\ x + 4y + 10z &= \lambda^2 \end{aligned}$ has a solution. Solve them completely in each case. | 05 | CO2 |
| (b) Define eigen value and eigen vector. Find the eigen value and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$. | 05 | CO2 |
| Q.6(a) Define exact differential equation. Test whether the equation $(x + y)^2 dx - (y^2 - 2xy - x^2) dy = 0$ is exact and hence solve it. | 04 | CO3 |
| (b) Form a differential equation of $y = a \ln x + b$. | 02 | CO3 |
| (c) Solve: $\frac{dy}{dx} = \frac{xy - 2y^2}{x^2 - 3xy}$. | 04 | CO3 |
| Q.7(a) Define an integrating factor. Explain it with an example. | 02 | CO3 |
| (b) Find an integrating factor and hence solve: $y \ln x dx + (x - \ln y) dy = 0$ | 04 | CO3 |
| (c) Identify and solve: $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$; $y(0) = 1$. | 04 | CO3 |
| Q.8 Solve the following differential equations: | 10 | CO3 |
| (a) $(D^2 + a^2)y = \tan ax$. | | |
| (b) $(D^3 - D^2 - 6D)y = x^2 + 1$. | | |
| (c) $(D^4 + D^2 + 1)y = ax^2 + be^{-x} \sin 2x$. | | |

Heaven's Light is Our Guide
Rajshahi University of Engineering & Technology
Department of Mechatronics Engineering

B.Sc. Engineering 1st Year Even Semester Examination 2024

Course No: Math 1227

Course Title: Vector, Matrix and Ordinary Differential Equation

Full Marks: 60

Time: Three Hours

N.B. i) Answer any Six questions, taking Three from each section.

ii) Figures in the margin indicate full marks.

iii) Use separate answer script for each section.

iv) CO-Course Outcome, PO-Program Outcome, BL- Bloom Taxonomy

SECTION-A

	Marks	CO
Q.1(a) Find an equation for the plane perpendicular to $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and passing through the terminal point to the vector $\vec{B} = \hat{i} + 5\hat{j} + 3\hat{k}$.	03	CO1
(b) Show that the vector $\vec{E} = \frac{\vec{r}}{r^2}$ is conservative. Find a scalar function ϕ such that $\vec{E} = -\nabla\phi$ and $\phi(a) = 0$, where $a > 0$.	03	CO1
(c) Evaluate $\int_C \vec{A} \cdot d\vec{r}$ along the curve $x^2 + y^2 = 1, z = 1$ in the positive direction of $(0,1,1)$ to $(1,0,1)$ if $\vec{A} = (yz + zx)\hat{i} + xz\hat{j} + (xy + 2z)\hat{k}$.	04	CO1
Q.2(a) Evaluate $\iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} \, ds$ where $\vec{F} = (x + 2y)\hat{i} - 3z\hat{j} + x\hat{k}$ and s is the surface of $2x + y + 2z = 6$ bounded by $x = 0, x = 1, y = 0, y = 2$.	05	CO1
(b) If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$ then evaluate $\iiint_V (\vec{\nabla} \times \vec{F}) \cdot d\vec{V}$ where V is the closed region bounded by the planes $x = 0, y = 0, z = 0, 2x + 2y + z = 4$.	05	CO1
Q.3(a) State and prove divergence theorem.	05	CO1
(b) Verify Stoke's theorem for the vector $\vec{F} = xz\hat{i} - y\hat{j} + x^2yz\hat{k}$ where s is the surface of the region bounded by $x = 0, y = 0, z = 0, 2x + y + 2z = 8$ which is not included in the xz plane.	05	CO1
Q.4(a) What is the adjoint of a matrix? Prove that the adjoint of a symmetric matrix is itself symmetric.	04	CO2
(b) Find the symmetric and skew symmetric part of the following matrix	03	CO2
$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$		
(c) Find the Hermitian part of the matrix	03	CO2
$A = \begin{bmatrix} 2 + 3i & 1 - i & 2 + i \\ 3 & 4 + 3i & 5 \\ 1 & 1 + i & 2i \end{bmatrix}$		

SECTION-B

Q.5(a) Find the eigen values and the corresponding eigen vectors of	05	CO2
$A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$		
(b) Solve by matrix method:	05	CO2
$\begin{aligned} x - y + 2z &= 3 \\ x + 2y + 3z &= 5 \\ 3x - 4y - 5z &= -13 \end{aligned}$		

Q.6(a) Define rank of a matrix. Reduce the following matrix

05 CO2

$$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix} \text{ into normal form and hence find its rank.}$$

(b) Define differential equation. Also mention the uses of differential equation.

02 CO3

(c) Solve: $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$.

03 CO3

Q.7(a) What is exact differential equation? Find the general solution of $(12y + 4x^3 + 6x^2)dx + 3(x + xy^2)dy = 0$.

03 CO3

(b) Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$.

03 CO3

(c) Find the integrating factor and hence find the general solution of $y \ln y dx + (x - \ln y)dy = 0$.

04 CO3

Q.8(a) Solve: $(x^3 D^3 + 2xD - 2)y = x^2 \ln x + 3x$.

05 CO3

(b) Find the current i from $L \frac{di}{dt} + Ri = E \sin 2t$, where L, R, E are constant subject to the condition $i = 0$ when $t = 0$.

05 CO4